DIFFERENTIABLE SATELLITE CONSTELLATION DESIGN VIA RELAXED COVERAGE AND REVISIT OBJECTIVES

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ABSTRACT

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1 Introduction

Designing satellite constellations is a fundamental problem in space mission design, across applications such as Earth Observation (EO), telecommunications, and navigation. The design space is high dimensional, with each spacecraft bringing up to six degrees of freedom corresponding to their orbital elements, and a wide range of often competing objectives to optimize over.

Traditional approaches address this complexity in one of two ways. Parametric families such as Walker Delta, Walker Star [20], and Flower constellations [14] reduce the design space to a fixed set of parameters but are inherently symmetric, making them non-optimal for non-uniform objectives. Metaheuristic methods such as genetic algorithms can handle arbitrary objectives but require thousands to millions of simulator rollouts [16].

For EO missions, the design problem is especially complex. Coverage and daytime revisit requirements may be non-uniform across ground targets, with additional constraints on data downlink. This non-uniformity makes parametric familie non-optimal, while the combinatorial nature of the problem makes metaheuristic methods computationally expensive.

Gradient-based optimization has also been previously attempted, but with limited success. Paek et al. [16] found the problem poorly conditioned for gradient-based methods, with difficulty even achieving convergence. Previous work has been limited the non-differentiability of the computational graph. Recent work by Acciarini et al. [2] has addressed part of this by developing ∂SGP4, an end-to-end differentiable SGP4 orbit propagator. However, the coverage and revisit metrics themselves remain non-differentiable, due to requiring binary visibility indicators, Boolean OR operations, and max functions.

To address this, we make the following contributions:

1. **Four continuous relaxations** that make coverage and revisit objectives differentiable: a soft sigmoid for visibility, a noisy-OR for multi-satellite aggregation, a leaky integrator for revisit gap tracking, and a LogSumExp soft-maximum for worst-case revisit (Figure 3).
2. **An end-to-end differentiable pipeline** from orbital elements to mission-level coverage metrics, composed with ∂SGP4 [2] and optimized with the AdamW optimizer (Figure 1).
3. **Demonstration on weighted target optimization**, where parametric families are inapplicable and prior gradient-based attempts failed.

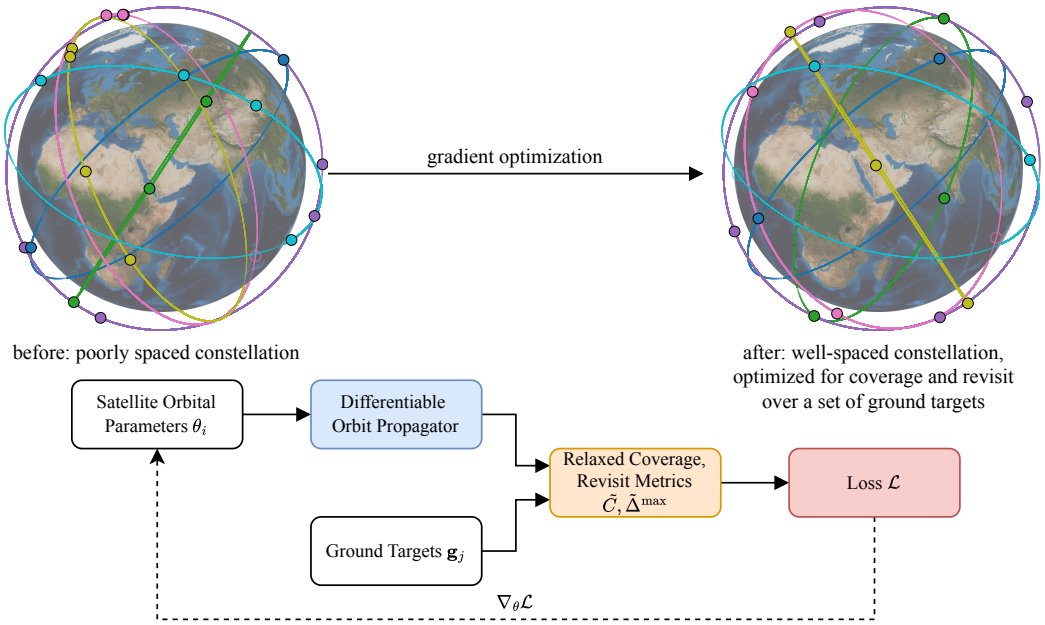


Figure 1: Overview of end-to-end differentiable constellation optimization. **Top:** Starting from a poorly-configured constellation with randomly spaced orbital planes and satellites, gradient optimization through the differentiable pipeline recovers a well-spaced constellation optimized for both coverage and revisit for a set of ground points. **Bottom:** Orbital parameters $\boldsymbol{\theta}_i$ are propagated through a differentiable orbit propagator and evaluated against a set of ground targets $\mathbf{g}_j$ to produce relaxed coverage and revisit metrics ($\tilde{C}$, $\tilde{\Delta}^{\max}$), which form a differentiable loss $\mathcal{L}$. Gradients $\nabla_{\boldsymbol{\theta}}\mathcal{L}$ flow back to the orbital parameters via reverse-mode automatic differentiation (dashed).

2 Related Work

2.1 Constellation Design and Optimization

Parametric constellations significantly reduce the number of design parameters, with the Walker Delta pattern [20] requiring only the number of planes, satellites per plane, and inclination. Flower constellations extend these ideas by using elliptical orbits to form specific 'harmonics' in revisit rates to produce repeating ground tracks [14], allowing for directed coverage over a specific area. These parametric families are efficient but inherently symmetric, and cannot be tailored to non-uniform or mission-specific coverage requirements without breaking the symmetry assumptions that make them tractable.

For non-parametric design, previous work has relied primarily on metaheuristic methods. Paek et al. [16] uses genetic algorithms and simulated annealing to reconfigure constellations, optimizing for global coverage and regional revisit over five design variables, requiring approximately 2500 total evaluations through expensive STK orbit simulations. They also attempted gradient-based optimization via finite differences and found it "extremely poorly conditioned," with convergence rarely achieved. In a survey of 144 papers on constellation design, Choo et al. [5] finds that gradient methods appear as a design approach in only two of the surveyed articles, making it among the least common methods—ranking far behind genetic algorithms, simulated annealing, and analytical approaches. The only gradient-based entry they identify, Fraire et al. [7], claims to be a gradient method, but in actuality uses a greedy neighbor search over discrete parameters such as number of satellites, planes, and inclination, with no analytic gradients computed. Abdelkhalik and Gad [1] uses a genetic algorithm to optimize single-orbit parameters for ground site coverage, explicitly noting that the fitness landscape contains numerous local minima that cause classical optimization methods to fail. Abdelkhalik and Gad [1] similarly claims to use a "second-order gradient" method, but no analytic gradients are used for optimization of design variables. Separately, Williams Rogers et al. [21] formulate constellation configuration as a collection of mixed-integer linear programs that select which satellites to activate from a pre-enumerated set of orbital slots, yielding provably optimal solutions for coverage and revisit objectives; however, the orbital slots and their underlying geometry (altitude, inclination, eccentricity) must be fixed a priori, so the method cannot discover improved orbit parameters.

Hwang et al. [10] demonstrates true gradient-based optimization of a satellite system through OpenMDAO, optimizing a single satellite's subsystem design across seven coupled disciplines with over 25,000 design variables, with gradient

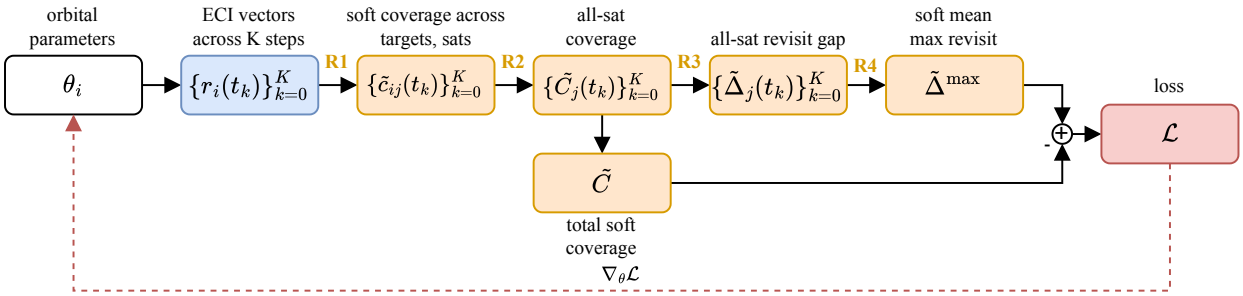


Figure 2: Computational graph for differentiable constellation optimization. Orbital parameters $\boldsymbol{\theta}_i$ are propagated to ECI position vectors $\{\mathbf{r}_i(t_k)\}$ across K timesteps. Four continuous relaxations (orange, **R1**–**R4**) transform the discrete coverage and revisit objectives into smooth surrogates: **R1** soft sigmoid visibility over targets and satellites, **R2** noisy-OR aggregation to all-satellite coverage, **R3** leaky integrator revisit gap accumulation, and **R4** LogSumExp soft-maximum over time. The coverage branch produces total soft coverage $\tilde{C}$; the revisit branch produces the soft mean worst-case revisit $\tilde{\Delta}^{\max}$. Both feed into the loss $\mathcal{L}$, and gradients $\nabla_{\boldsymbol{\theta}}\mathcal{L}$ flow back via reverse-mode automatic differentiation (dashed).

computation made tractable with adjoint state methods. Their work utilizes gradient-based objectives combined with orbit-dependent parameters, such as solar panel angles, battery sizing, communication scheduling, to optimize a single spacecraft's operation.

2.2 Differentiable Physics and Simulation

Differentiable programming [3] allows for gradients to be obtained across domains even where forward models are complex but smooth. The benefits of differentiable programming have been made significantly more accessible through frameworks that perform automatic differentiation, such as PyTorch and JAX. Hu et al. [9] demonstrated differentiable physical simulators targeting robotics and fluid dynamics, and de Avila Belbute-Peres et al. [6] develops differentiable physics engines to contact-rich robotics. For continuous relaxations applied to parameterized categorical distributions, Gumbel-Softmax reparameterization [11] is frequently used to compute analytic gradients. Kamra and Liu [12] proves a coverage gradient theorem for spatial multi-resource coverage objectives, providing an estimator for the gradient that can be used for optimization of sensor placement, via spatial discretization and implicit boundary differentiation applied to 2D planar sensor networks.

In astrodynamics, Acciarini et al. [2] introduces ∂SGP4, a PyTorch reimplementation of the SGP4/SDP4 orbit propagation algorithm, which is forward mode differentiable. Naylor et al. [15] extends this direction by combining a JAX-based differentiable SGP4 propagator with Mitsuba , a differentiable renderer, to optimize trajectories for spacecraft-to-spacecraft inspection operations, demonstrating end-to-end forward-mode differentiation through orbit propagation chained with a rendering pipeline. cite

This work builds on ∂SGP4 by introducing differentiable coverage and revisit objectives that enable direct optimization of mission performance metrics.

3 Method

3.1 Problem Formulation

We consider a constellation of N satellites, each parameterized by a two-line element (TLE) set. Each satellite brings up to a maximum of six degree of freedom corresponding to their orbital elements, $[a,\ e,\ i,\ \Omega,\ \omega,\ \mathcal{M}]^\top$, corresponding to semi-major axis, eccentricity, inclination, right ascension of the ascending node (RAAN), argument of periapsis, and mean anomaly, respectively. We universally fix the remaining TLE parameters corresponding to mean motion n, drag term B^*, and secular rates $\dot{n}$, $\ddot{n}$ to keep the constellation configuration static over time.

Given a grid of J ground target points $\mathcal{G} = \{\mathbf{g}_1, \ldots, \mathbf{g}_J\}$ on the Earth's surface and a time horizon $[0, T]$ discretized into K steps with spacing $\Delta t = T/(K-1)$, we seek to maximize coverage over the Earth and minimize the worst-case revisit gap.

Propagation. Each satellite's TLE elements are propagated through ∂SGP4 [2] to produce position vectors $\mathbf{r}_i(t_k)$ in the True Equator Mean Equinox (TEME) reference frame. These are rotated to the Earth-Centered Earth-Fixed (ECEF) frame via the Greenwich Mean Sidereal Time angle $\theta_{\text{GMST}}(t_k)$:

$$\mathbf{r}_i^{\text{ECEF}}(t_k) = \mathbf{R}_z\left(\theta_{\text{GMST}}(t_k)\right)\, \mathbf{r}_i^{\text{TEME}}(t_k), \tag{1}$$

where $\mathbf{R}_z(\theta)$ is the standard rotation matrix about the z-axis. This single-axis rotation is an approximation and neglects polar motion; the complete TEME-to-ECEF transformation requires an additional rotation between the Earth's instantaneous rotation axis and the ECEF pole [19]. For the $\sim 24\,\text{h}$ propagation horizons considered in this work, polar-motion offsets are on the order of tenths of arcseconds, corresponding to sub-kilometre position errors. These errors are small relative to the $\sim 2200\,\text{km}$ ground footprint of a satellite at $550\,\text{km}$ altitude with 10° minimum elevation.

Elevation angle. The elevation angle from ground point $\mathbf{g}_j$ to satellite i at time t_k is

$$\alpha_{ij}(t_k) = \arcsin\left(\frac{(\mathbf{r}_i^{\text{ECEF}}(t_k) - \mathbf{g}_j)\cdot \hat{\mathbf{g}}_j}{\|\mathbf{r}_i^{\text{ECEF}}(t_k) - \mathbf{g}_j\|}\right), \tag{2}$$

where $\hat{\mathbf{g}}_j = \mathbf{g}_j/\|\mathbf{g}_j\|$ is the local vertical unit vector at the ground point. A satellite is visible when $\alpha_{ij} \geq \alpha_{\min}$, where we consider the minimum elevation angle $\alpha_{\min} = 10^\circ$, which varies depending on mission requirements.

Coverage and revisit. Instantaneous coverage at ground point $\mathbf{g}_j$ at time t_k is the indicator function for when at least one satellite is visible:

$$C_j(t_k) = \mathbf{1}\left[\bigvee_{i=1}^{N} \alpha_{ij}(t_k) \geq \alpha_{\min}\right]. \tag{3}$$

The instantaneous coverage fraction over the full set of ground points $\mathcal{G}$ is then

$$C = \frac{1}{KJ}\sum_{k=1}^{K}\sum_{j=1}^{J} C_j(t_k). \tag{4}$$

The revisit gap at ground point $\mathbf{g}_j$ is the elapsed time since the most recent coverage event:

$$\Delta_j(t_k) = t_k - t_{k'}, \quad k' = \max\left\{k'' \leq k : C_j(t_{k''}) = 1\right\} \tag{5}$$

and the mean worst-case revisit across all ground points is then

$$\Delta^{\max} = \frac{1}{J}\sum_{j=1}^{J} \max_k \Delta_j(t_k). \tag{6}$$

We now introduce the relaxations required to make these metrics differentiable.

3.2 Relaxation 1: Soft Visibility

We replace the hard step function with a sigmoid (Figure 3a):

$$\tilde{c}_{ij}(t_k) = \sigma\left(\frac{\alpha_{ij}(t_k) - \alpha_{\min}}{\tau}\right), \quad \sigma(x) = \frac{1}{1+e^{-x}}, \tag{7}$$

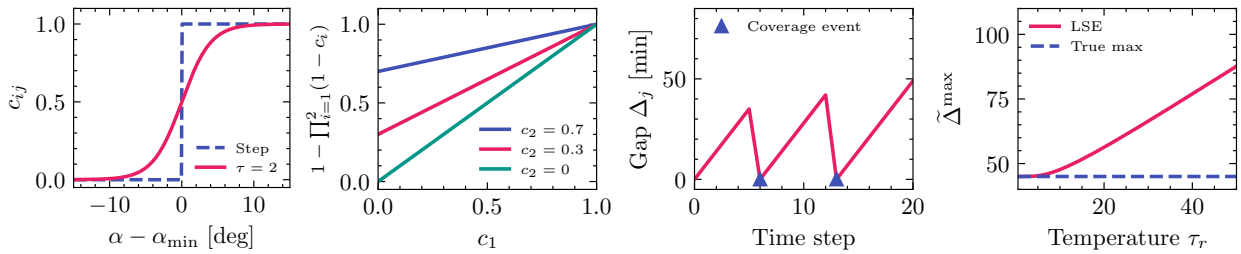
where $\tau > 0$ controls the sharpness. As $\tau \to 0$ this recovers the hard indicator. This relaxation is analogous to the temperature-scaled softmax used in Gumbel-Softmax relaxations of discrete variables [11].

3.3 Relaxation 2: Noisy-OR Coverage Aggregation

Starting with the Boolean OR operator in Equation 3, treating each $\tilde{c}_{ij}(t_k)$ as an independent detection probability, the noisy-OR [18] relaxation gives (Figure 3b):

$$\tilde{C}_j(t_k) = 1 - \prod_{i=1}^{N}\left(1 - \tilde{c}_{ij}(t_k)\right). \tag{8}$$

This function is smooth, monotonically increasing in each $\tilde{c}_{ij}$, and recovers the hard OR when all inputs are in $\{0, 1\}$. This relaxation also inherently handles overlapping coverage: additional coverage of a point produces diminishing marginal gain, preventing the optimizer from clustering satellites.



(a) Soft sigmoid (b) Noisy-OR (c) Leaky integrator (d) LogSumExp

Figure 3: The four continuous relaxations used in this work to produce a differentiable loss landscape.

The soft coverage fraction over the full set of ground points $\mathcal{G}$ with soft visibility and noisy-OR aggregation then becomes

$$\tilde{C} = \frac{1}{KJ} \sum_{k=1}^{K} \sum_{j=1}^{J} \tilde{C}_j(t_k). \tag{9}$$

3.4 Relaxation 3: Leaky Integrator for Revisit Gaps

The revisit gap at a ground point is the time elapsed since the most recent coverage event. Computing this gap requires identifying discrete transitions which we instead approximate with a leaky integrator recurrence [8] (Figure 3c):

$$\tilde{\Delta}_j(t_k) = \left(\tilde{\Delta}_j(t_{k-1}) + \delta t\right) \cdot \left(1 - \tilde{C}_j(t_k)\right), \quad \tilde{\Delta}_j(t_0) = 0. \tag{10}$$

When the ground point $\mathbf{g}_j$ is covered ($\tilde{C}_j \approx 1$), the accumulated gap is multiplied by approximately zero, resetting it. When the point is not covered ($\tilde{C}_j \approx 0$), the gap grows by Δt. This recurrence is fully differentiable and its maximum over time approximates the true worst-case revisit gap at each ground point.

3.5 Relaxation 4: LogSumExp Soft-Maximum

The soft worst-case revisit gap at ground point $\mathbf{g}_j$ is $\max_k \tilde{\Delta}_j(t_k)$, which has zero gradients almost everywhere. We replace it with the LogSumExp smooth maximum [4] (Figure 3d):

$$\widetilde{\Delta}_j^{\max} = \beta \cdot \log\left(\sum_{k=1}^{K} \exp\left(\frac{\tilde{\Delta}_j(t_k)}{\beta}\right)\right), \tag{11}$$

where β is a sharpness parameter, for which we assign $\beta = 30\,\text{min}$. This relaxation satisfies the inequality $\max_k \tilde{\Delta}_j(t_k) \leq \widetilde{\Delta}_j^{\max} \leq \max_k \tilde{\Delta}_j(t_k) + \beta \log K$, providing an upper bound. The mean worst-case revisit across all ground points is then

$$\tilde{\Delta}^{\max} = \frac{1}{J} \sum_{j=1}^{J} \widetilde{\Delta}_j^{\max}. \tag{12}$$

In practice, the coverage and revisit objectives have different sensitivity to the sigmoid temperature τ in Equation 7: phantom coverage is an additive bias for coverage but multiplicatively resets the revisit integrator. We therefore split τ into τ_{cov} for the coverage objective and τ_{rev} for the revisit computation, calibrated in §4.2.

3.6 Combined Objective

The total loss combines coverage maximization with revisit minimization:

$$\mathcal{L}(\mathbf{x}) = -\tilde{C} + \lambda\, \tilde{\Delta}^{\max}, \tag{13}$$

where $\mathbf{x}$ collects all free and reparameterized (§3.8) orbital parameters and λ is a weighting coefficient. The coverage term is instantaneous: it penalizes overlapping fields of view and saturates once satellites are well-separated. The revisit term, aggregated over the full simulation, continues providing gradients after coverage plateaus. Although both terms

push toward spacing satellites apart, they operate at different scales—coverage quickly separates clusters, while revisit works globally.

For objectives that are not geodesically uniform, the sums over ground points can be replaced with weighted sums over a discrete target set (§4.5).

3.7 Tightness of the Composed Relaxation

Each of the four relaxations is individually tight: the soft sigmoid recovers the hard step as $\tau \to 0$; the noisy-OR is exact when its inputs are binary; the leaky integrator is exact when coverage is in $\{0, 1\}$; and the LogSumExp satisfies $\max_k x_k \leq \text{LSE} \leq \max_k x_k + \beta \log K$. However, the *composition* of these relaxations does not inherit these guarantees, because each relaxation receives the soft output of the previous one rather than the binary values for which it was designed.

The interaction introduces two competing biases:

Phantom coverage (gap shortening). The soft sigmoid coverage indicator assigns nonzero coverage $\tilde{c}_{ij} > 0$ to satellites slightly *below* the visibility threshold ($\alpha_{ij} < \alpha_{\min}$). This phantom coverage propagates through the noisy-OR and partially resets the leaky integrator, producing gaps that are *shorter* than the true discrete gaps.

Incomplete reset (gap lengthening). Conversely, a satellite well above the threshold produces $\tilde{c}_{ij} \approx 0.99$, not exactly 1. The leaky integrator retains a fraction $(1 - \tilde{C}_j) \approx 0.01$ of the accumulated gap at each coverage event. Over time, this residual prevents the gap from fully resetting, producing relaxed gaps that are *longer* than the true gaps during sustained coverage. The LogSumExp adds a further upward bias of at most $\beta \log K$.

These effects compete: phantom coverage shortens gaps near the visibility boundary, while incomplete resets and the LogSumExp overestimate lengthen them. The net bias depends on the specific geometry and is neither a guaranteed upper nor lower bound on the true objective. The asymmetric sensitivity—additive for coverage, multiplicative for revisit—motivates the split sigmoid temperature introduced in §3.5.

The relaxation temperatures control a tradeoff: smaller values tighten the approximation but sharpen the gradients, potentially destabilizing optimization. We investigate this tradeoff empirically in §4.2 by computing the hard coverage and revisit metrics at the optimizer's solution and comparing them to the relaxed values, as well as through PCA over the optimizer landscape (Figure 6).

3.8 Optimization

Computational graph. The full forward pass composes the ∂SGP4 propagator with the four relaxations into a single differentiable graph (Figure 2). Every operation is implemented in PyTorch [17], so $\nabla_{\theta_i} \mathcal{L}$ is computed via reverse-mode automatic differentiation through the entire graph. The Jacobian $\partial \mathbf{r}_i^{\text{TEME}} / \partial \mathbf{x}_i$ is handled inside ∂SGP4: it is dense and time-varying—a small change in RAAN rotates the entire orbital plane, affecting position at all future time steps.

Interval constraints and reparameterization. Without constraints, the optimizer trivially raises altitude until all ground points are permanently visible. We enforce interval constraints (e.g. LEO altitude bounds) via sigmoid reparameterization, mapping unconstrained variables θ_i to valid elements $\mathbf{x}_i$:

$$x_d = \ell_d + (u_d - \ell_d)\, \sigma(\theta_d) \quad \Rightarrow \quad \frac{\partial x_d}{\partial \theta_d} = (u_d - \ell_d)\, \sigma(\theta_d)(1 - \sigma(\theta_d)), \tag{14}$$

for bounded parameters (e.g. inclination) with bounds $[\ell_d, u_d]$, and $x_d = \theta_d$ (identity) for periodic parameters (RAAN, mean anomaly) where wrapping is natural. This reparameterization eliminates the need for projected gradient or constrained optimization methods.

Plane constraints via gradient accumulation. In many constellation designs, satellites within the same orbital plane share a common RAAN while maintaining individual mean anomalies. Rather than enforcing this as a hard constraint, we average these gradients per-plane: if satellites $i_1, \ldots, i_S$ in plane p share a single RAAN parameter Ω_p, the gradient is the average of per-satellite contributions:

$$\frac{\partial \mathcal{L}}{\partial \Omega_p} = \frac{1}{S} \sum_{s=1}^{S} \frac{\partial \mathcal{L}}{\partial \Omega_{i_s}}, \tag{15}$$

The averaged gradient is applied to the shared parameter, and the updated value is synchronized across all satellites in the plane after each optimizer step. This reduces the effective dimensionality without modifying the forward pass, and generalizes to any parameter-tying pattern: per-plane inclination sharing, altitude coupling across the constellation, or any grouping governed by mission constraints.

4 Experiments

We first calibrate the relaxation hyperparameters to ensure the surrogate loss faithfully ranks candidate solutions. We then evaluate the framework on three experiments: a toy problem validating gradient correctness, a uniform-coverage baseline that recovers known-optimal geometries, and a weighted target optimization on a problem class where parametric families fail.

4.1 Common Setup

All experiments use dSGP4 [2] for orbit propagation with PyTorch autograd for gradient computation. Circular orbits are assumed ($e = 0.001$) at $550\,\mathrm{km}$ altitude, held fixed by zeroing the mean-motion gradient. The minimum elevation angle is $\alpha_{\min} = 10°$. As described in §4.2, we use separate sigmoid temperatures for coverage and revisit: the coverage temperature τ_{cov} controls gradient smoothness, while a tighter revisit temperature τ_{rev} prevents the leaky integrator from being gamed by phantom coverage. The LogSumExp temperature is β. Coverage and revisit metrics are evaluated over a $24\,\mathrm{h}$ propagation horizon discretized into $K = 240$ time steps on a ground grid of $36 \times 72 = 2592$ points spanning $\pm 70°$ latitude. Specific relaxation parameters, optimizer settings, and revisit weight λ vary per experiment and are stated in each subsection.

4.2 Hyperparameter Tuning

The composed relaxation has four hyperparameters that define the loss surface geometry: the coverage sigmoid temperature τ_{cov}, the revisit sigmoid temperature τ_{rev} (introduced in §3.2), the LogSumExp temperature β, and the revisit weight λ. We exclude optimizer dynamics (learning rate, momentum) from this calibration, as they affect convergence speed but not the shape of the loss landscape. We calibrate by running four optimizations of a 24-satellite constellation (6×4, inclination fixed at $60°$, RAAN and mean anomaly free, per-plane RAAN sharing) from different initial RAAN configurations:

1. **Near-uniform**: $\Omega = [0, 60, 120, 180, 240, 300]°$ (close to Walker spacing),
2. **Moderate**: $\Omega = [0, 30, 120, 200, 210, 300]°$ (irregular but spread),
3. **Clustered**: $\Omega = [0, 10, 20, 30, 40, 50]°$ (all planes within $50°$),
4. **Two-cluster**: $\Omega = [0, 5, 180, 185, 270, 275]°$ (three tight clusters).

All runs use the same random mean anomaly initialization (seed 42) and default relaxation parameters ($\tau = 2°$, $\beta = 10\,\mathrm{min}$, $\lambda = 0.1$) for 800 Adam iterations.

Ranking failure with shared softness. The near-uniform and moderate initializations converge to solutions with hard mean revisit gaps of $\sim 48\,\mathrm{min}$, while the clustered and two-cluster initializations find solutions with gaps of $\sim 68\,\mathrm{min}$. However, under the default relaxation parameters, the *relaxed* loss of the $68\,\mathrm{min}$ solutions is *lower* than that of the $48\,\mathrm{min}$ solutions—the relaxation ranks the worse solutions as better.

This ranking failure is caused by the shared sigmoid temperature: the same τ controls both the coverage objective and the revisit leaky integrator (Equation 10). A soft sigmoid with $\tau = 2°$ assigns coverage $c_{ij} \approx 0.27$ to a satellite at $\alpha = \alpha_{\min} - 2°$. For the coverage fraction (Equation 9), this phantom coverage has minimal effect because it is a small additive bias. But in the leaky integrator, it *multiplicatively* resets the gap: $\Delta_j \leftarrow \Delta_j \cdot (1 - C_j)$. When multiple clustered satellites each contribute partial coverage at low elevation, the noisy-OR combines them into substantial phantom coverage ($C_j \gg 0$) that partially resets the gap timer at every time step. The clustered solutions benefit disproportionately from this effect.

Split-softness resolution. We decouple the sigmoid temperature into τ_{cov} for the coverage objective and τ_{rev} for the revisit integrator, then grid-search over $\tau_{\text{cov}} \in \{1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0\}°$, $\tau_{\text{rev}} \in \{0.1, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0\}°$, $\beta \in \{3, 5, 7.5, 10, 15, 20\}\mathrm{min}$, and $\lambda \in \{0.05, 0.1, 0.2, 0.5, 1.0, 2.0\}$ (1764 combinations, each requiring only four gradient-free forward passes). A combination is *valid* if it correctly ranks the good solutions (near-uniform and moderate) below the bad solutions (clustered and two-cluster).

Table 1: Split-softness calibration results. Valid = relaxed loss correctly ranks solutions with better hard metrics below those with worse hard metrics. Decoupling the coverage and revisit sigmoid temperatures allows smooth coverage gradients with a tight revisit sigmoid.

Regime	**Softness range**	**Valid combos**
Shared $\tau_{\text{cov}} = \tau_{\text{rev}} \geq 2°$	Original setting	0
Shared $\tau_{\text{cov}} = \tau_{\text{rev}} = 1°$	Tighter, sharp gradients	Some (with $\beta \leq 5$, $\lambda \geq 0.2$)
Split: $\tau_{\text{cov}} \geq 2°$, $\tau_{\text{rev}} \leq 0.5°$	Smooth + tight	Many
Split: $\tau_{\text{cov}} \geq 3°$, $\tau_{\text{rev}} \leq 1°$	Recommended	Many

Table 1 summarizes the findings. No valid combinations exist when $\tau_{\text{cov}} = \tau_{\text{rev}} \geq 2°$ (the original shared-softness regime). Decoupling the temperatures resolves the ranking failure: combinations with $\tau_{\text{cov}} \geq 3°$ and $\tau_{\text{rev}} \leq 0.5°$ correctly rank solutions while maintaining smooth coverage gradients. See Appendix A for the full grid search results.

Smoothness–correctness tradeoff. The valid hyperparameter combinations tend toward tight revisit sigmoids ($\tau_{\text{rev}} \leq 1°$), which produce correct global rankings but sharp, poorly conditioned gradients. In practice, optimization converges more reliably with softer values (e.g., $\tau_{\text{cov}} = 5°$, $\tau_{\text{rev}} = 2°$) that lie outside the valid regime. This is not contradictory: the calibration tests *global* ranking correctness across qualitatively different solutions, but gradient-based optimization only requires *local* gradient quality within a basin of attraction. Softer relaxations smooth the loss surface, improving convergence, while the underlying basin structure still guides the optimizer toward good solutions. The hyperparameters used in subsequent experiments reflect this tradeoff, favoring reliable convergence over exact global ranking.

4.3 Experiment 1: Toy Problem

To validate that the relaxed objectives produce correct gradients and results, we consider the simplest non-trivial constellation design problem: two satellites in a single orbital plane at $550\,\text{km}$ altitude with inclination fixed at $60°$ and RAAN fixed at $0°$, optimizing only their mean anomalies $\mathcal{M}_1$ and $\mathcal{M}_2$. We use relaxation parameters $\tau_{\text{cov}} = 5°$, $\tau_{\text{rev}} = 2°$, $\beta = 10\,\text{min}$, $\lambda = 2.0$, and AdamW with learning rate $\eta = 10^{-2}$.

The optimal solution is known analytically: the two satellites should be separated by $180°$ in mean anomaly to maximize coverage and revisit rate within the shared orbital plane. This reduces the problem to two free parameters, allowing the full loss landscape to be visualized as a 2D contour plot.

Loss landscape. We evaluate both the relaxed loss and hard (discrete) metrics on a dense grid over $(\mathcal{M}_1, \mathcal{M}_2) \in [0, 360)^{2\circ}$. Figure 4 shows the resulting landscapes: the top row displays the relaxed loss, coverage, and revisit, while the bottom row shows the corresponding hard metrics. Both landscapes exhibit a clear global minimum along the anti-diagonal $|\mathcal{M}_1 - \mathcal{M}_2| = 180°$, with no spurious local minima. The relaxed landscape is visibly smoother than the hard landscape while preserving the same basin structure, confirming that the relaxations provide a smooth surface amenable to gradient descent without distorting the underlying objective.

Gradient descent trajectory. Starting from $\mathcal{M}_1 = 120°$, $\mathcal{M}_2 = 130°$ (a $10°$ separation), Adam converges to $|\mathcal{M}_1 - \mathcal{M}_2| \approx 180°$ within 400 iterations. The trajectory (overlaid in white on Figure 4) follows the gradient field smoothly toward the minimum without oscillation. This validates that the composed relaxations produce well-behaved gradients—a prerequisite for the higher-dimensional experiments that follow.

4.4 Experiment 2: Uniform Coverage Baseline

Before applying the method to non-uniform objectives, we establish a baseline on the well-understood uniform coverage problem. This validates that the optimizer can recover known-optimal geometries and provides a point of comparison for the weighted-target experiment.

We optimize $N = 24$ satellites in 6 orbital planes (4 per plane) with inclination fixed at $60°$. RAAN and mean anomaly are free, with RAAN shared within each plane, yielding $6 + 24 = 30$ effective degrees of freedom. We use AdamW with learning rate $\eta = 10^{-2}$ and revisit weight $\lambda = 0.1$.

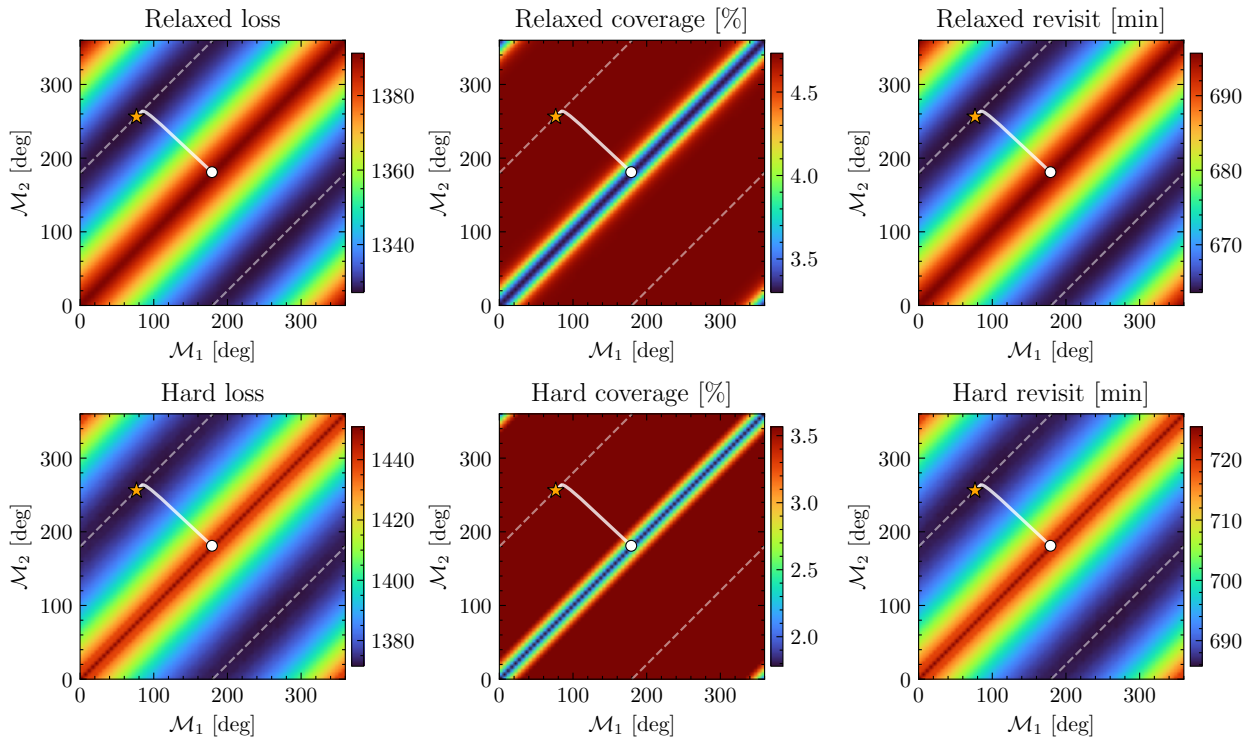


Figure 4: Loss landscape for the two-satellite mean anomaly optimization. **Top row:** relaxed loss, coverage, and revisit. **Bottom row:** corresponding hard (discrete) metrics. The global minimum lies along $|\mathcal{M}_1 - \mathcal{M}_2| = 180°$ (dashed line). The gradient descent trajectory (white line, circle = start, star = end) converges from a $10°$ separation to the optimum. The relaxed landscape is smoother than the hard landscape while preserving the same basin structure.

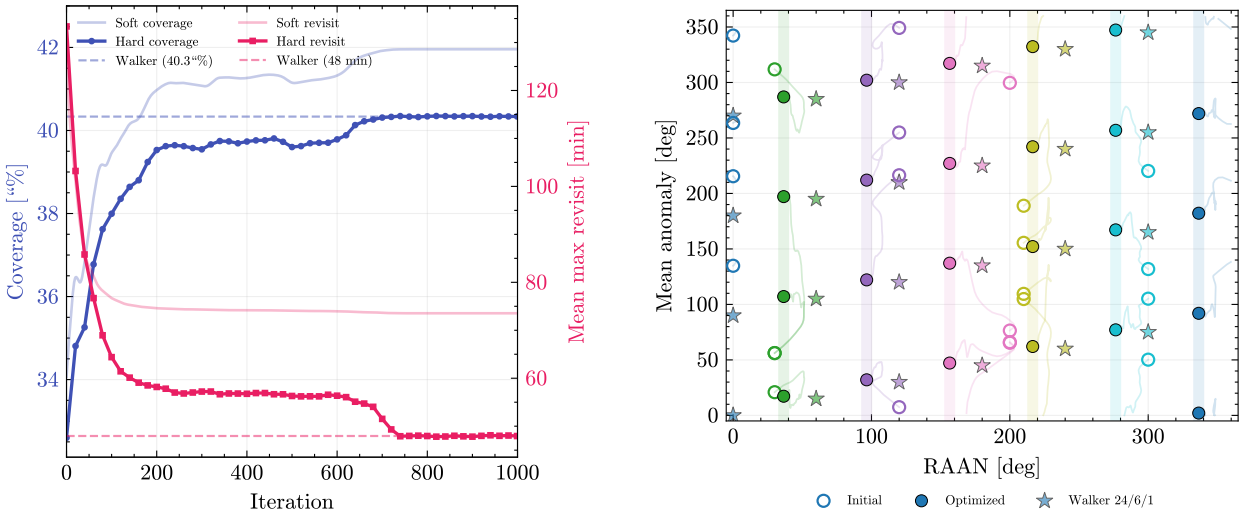
Table 2: Walker recovery: 24-satellite constellation (6×4), 1000 iterations, $550\,\mathrm{km}$ fixed altitude, inclination fixed at $60°$. Soft metrics are the relaxed surrogate values; hard metrics are the discrete ground truth.

Metric	**Initial**	**Optimized**	**Walker 24/6/1**
Hard coverage [%]	32.41	40.34	40.33
Hard mean max revisit [min]	136.2	48.0	48.0
Soft coverage [%]	41.86	41.96	50.32
Soft mean max revisit [min]	127.5	73.5	73.5

Initial configuration. We deliberately initialize the constellation with irregular RAAN offsets $[0, 30, 120, 200, 210, 300]°$ (far from uniform $60°$ spacing) and random mean anomalies (seed 42). The classical Walker-delta $24/6/1$ pattern uses $60°$ RAAN spacing.

Results. Over 1000 iterations, the optimizer recovers near-uniform RAAN spacing from the irregular initialization, approaching the Walker-delta geometry without domain-specific heuristics. Table 2 summarizes the metrics; Figure 5 shows convergence, spatial coverage redistribution, and ground tracks.

Loss landscape. Following Li et al. [13], we visualize the loss landscape by computing per-trajectory principal components and evaluating the landscape on each trajectory's own 2D slice through the 30-dimensional parameter space. Unlike a single global PCA projection—which discards the dimensions that matter for individual trajectories—each slice faithfully represents the landscape that its optimizer actually navigated. Figure 6 shows smooth, well-conditioned basins across all four initializations, in contrast to the "extremely poorly conditioned" landscape reported by Paek et al. [16] when using finite-difference gradients through discrete objectives. This confirms that the poor conditioning observed in prior work was a consequence of the non-differentiable objective formulation, not the underlying physics.



(a) Convergence

(b) RAAN vs mean anomaly

Figure 5: Walker recovery experiment. **(a)** Convergence of soft and hard coverage and revisit metrics over 1000 iterations, with Walker 24/6/1 reference (dashed). The optimizer matches Walker performance by iteration ∼400. **(b)** RAAN vs mean anomaly trajectories. Hollow circles show the initial configuration, solid circles the optimized result, and stars the Walker reference.

4.5 Experiment 3: Regional Target Optimization

The primary advantage of gradient-based constellation optimization over parametric families is the ability to handle non-uniform, mission-specific objectives. We demonstrate this by optimizing a small constellation for coverage and revisit over Europe—a problem that Walker constellations cannot address by construction.

Setup. We sample 500 ground targets from European country boundaries (GeoJSON), weighted by $\cos(\text{lat})$ to correct for area distortion. The constellation consists of 4 satellites in 2 orbital planes (2 per plane). Unlike Experiment 2, we free all orbital geometry parameters: inclination ($\iota \in [30, 90]^\circ$), RAAN, mean anomaly, argument of perigee, and orbital shape. To allow the optimizer to explore eccentric orbits without producing unphysical perigee altitudes, we reparameterize eccentricity and altitude jointly as (perigee altitude, excess altitude), where perigee $\in [400, 600]\,\text{km}$ and excess $\in [0, 1500]\,\text{km}$, guaranteeing perigee $\geq 400\,\text{km}$ by construction. Inclination, RAAN, eccentricity, argument of perigee, and altitude are shared within each plane; mean anomaly is free per satellite, giving 12 effective degrees of freedom. We use AdamW with $\eta = 10^{-2}$, $\lambda = 1.0$, and run for 2000 iterations.

Results. Starting from a Walker-like circular initialization at $550\,\text{km}$, the optimizer discovers a qualitatively different geometry (Table 3). Both planes converge to eccentric orbits ($e \approx 0.09$) with argument of perigee $\omega \approx 270^\circ$, placing apogee over the northern hemisphere where the satellite dwells longest. The resulting orbits have perigee $\approx 580\,\text{km}$ and apogee $\approx 1960\,\text{km}$—a Molniya-like dwell strategy discovered entirely from gradients, with no domain-specific heuristics. The inclination converges to $\sim 64^\circ$, matching European latitudes.

The optimized constellation achieves 38.2% coverage and 41.9 min mean revisit over Europe, compared to 10.0% coverage and 247.5 min for a Walker 4/2/1 baseline—a $4\times$ improvement in coverage and $6\times$ in revisit with the same number of satellites.

Figure 8 shows the visibility density (number of timesteps each ground bin is within 10° elevation of any satellite). The optimized constellation concentrates coverage in the northern hemisphere over Europe, while the Walker baseline distributes coverage symmetrically. The density map also reveals longitudinal structure: the orbital period of $\sim 111\,\text{min}$ produces approximately 13 revolutions per sidereal day, creating a near-resonant ground track that clusters revisits at preferred longitudes.

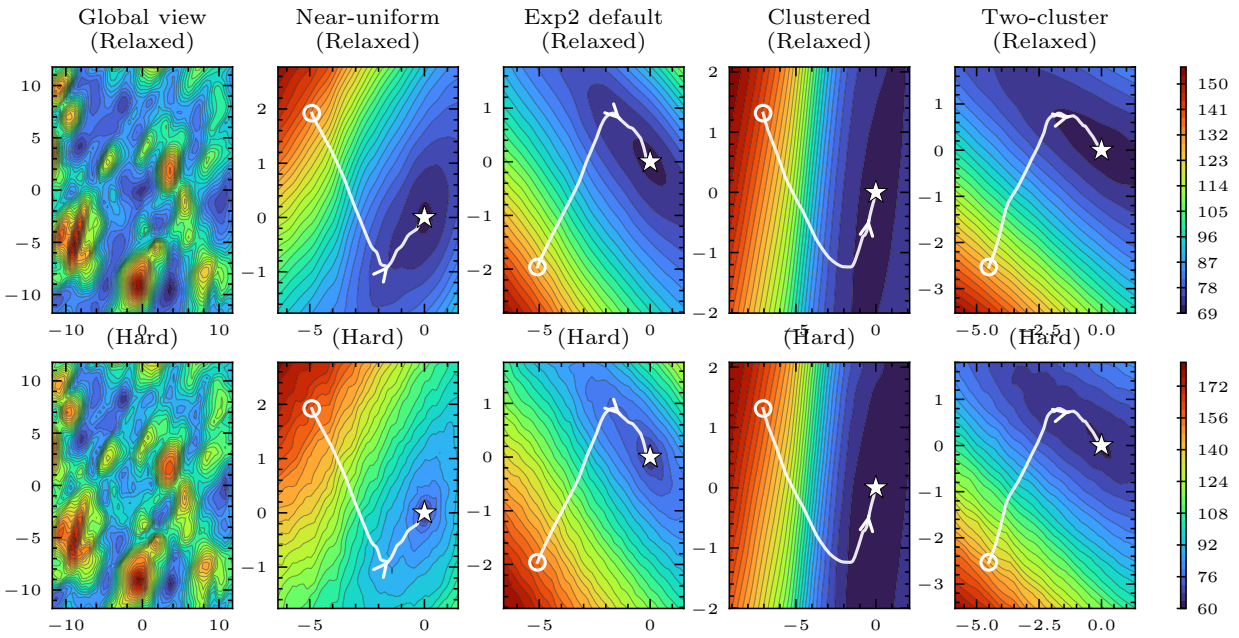


Figure 6: Loss landscape visualization [13]. **Left column:** global view along random directions (zoomed out) revealing the broader landscape structure. **Columns 2–5:** per-trajectory PCA slices, each showing the landscape along that trajectory's own principal components (white line, circle = start, star = end). **Top row:** relaxed total loss. **Bottom row:** hard (discrete) total loss. The per-trajectory views show smooth basins, while the global view reveals the multi-modal structure of the full parameter space.

Table 3: Regional target optimization: 4 satellites (2 planes $\times$ 2), 2000 iterations, all orbital geometry parameters free with coupled perigee/apoapsis constraints.

Metric	**Walker 4/2/1**	**Optimized**
Hard coverage [%]	9.97	38.20
Hard mean max revisit [min]	247.5	41.9
Soft coverage [%]	12.37	40.31
Soft mean max revisit [min]	208.9	75.1

5 Conclusions and Future Work

Gradient-based optimization of satellite constellations has been considered intractable due to the discrete, non-differentiable nature of coverage and revisit objectives. We showed that this intractability is an artifact of the objective formulation: by introducing four continuous relaxations and composing them with the differentiable ∂SGP4 orbit propagator, the entire pipeline becomes amenable to first-order optimization. The relaxed surrogate closely tracks the discrete objective (§4.2) and produces well-conditioned gradients where finite-difference approaches through discrete objectives [16] do not.

On uniform coverage, the optimizer recovers Walker-optimal geometry from arbitrary initialization. On weighted regional targets, it discovers asymmetric constellations—including eccentric orbits that dwell over target latitudes—that significantly outperform Walker baselines, without any domain-specific heuristics.

Limitations. The current approach optimizes a single-epoch snapshot and does not account for long-term perturbations (J2 RAAN drift, atmospheric drag) that alter constellation geometry over months. The time step discretization introduces approximation error in the revisit metrics. While the loss landscape appears smooth near our solutions, we cannot rule out the existence of better local minima elsewhere in the parameter space. The discrete design variables (number of planes, satellites per plane) remain non-differentiable and must be set by the user or searched over separately.

Table 4: Per-plane orbital geometry discovered by the optimizer for the Europe target set. Both planes converge to eccentric, Molniya-like orbits with apogee over the northern hemisphere ($\omega \approx 270°$). The Walker baseline uses circular orbits at $550\,\mathrm{km}$.

	ι [deg]	Ω [deg]	e	ω [deg]	Perigee / apogee [km]
Plane 0	63.5	343.7	0.090	269.4	580 / 1960
Plane 1	63.6	161.0	0.090	272.5	580 / 1960
Walker (all)	60.0	uniform	0.001	—	550 / 550

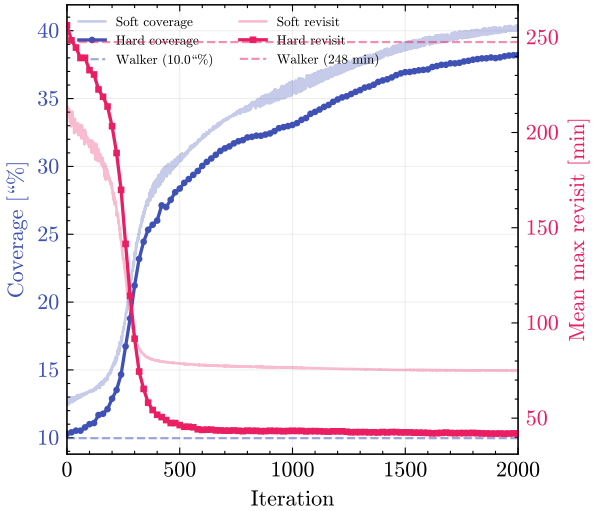


Figure 7: Convergence of the Europe target optimization over 2000 iterations. The optimizer rapidly improves both coverage and revisit over the Walker baseline (dashed).

Future work. Incorporating J2 secular perturbations would enable optimization of long-term constellation maintenance, potentially discovering geometries that exploit RAAN drift for improved coverage evolution. Multi-objective Pareto optimization would allow explicit trade-offs between competing mission requirements. More broadly, the framework extends to constellation augmentation—optimizing new satellites to complement an existing fleet—a setting that completely breaks the symmetry assumptions of parametric families.

References

[1] Ossama Abdelkhalik and Ahmed Gad. Optimization of space orbits design for Earth orbiting missions. *Acta Astronautica*, 68:1307–1317, 2011.

[2] Giacomo Acciarini, Atılım Güneş Baydin, and Dario Izzo. Differentiable orbital mechanics with automatic differentiation. *Acta Astronautica*, 225:679–687, 2025.

[3] Atılım Güneş Baydin, Barak A. Pearlmutter, Alexey Andreyevich Radul, and Jeffrey Mark Siskind. Automatic differentiation in machine learning: A survey. *Journal of Machine Learning Research*, 18(153):1–43, 2018.

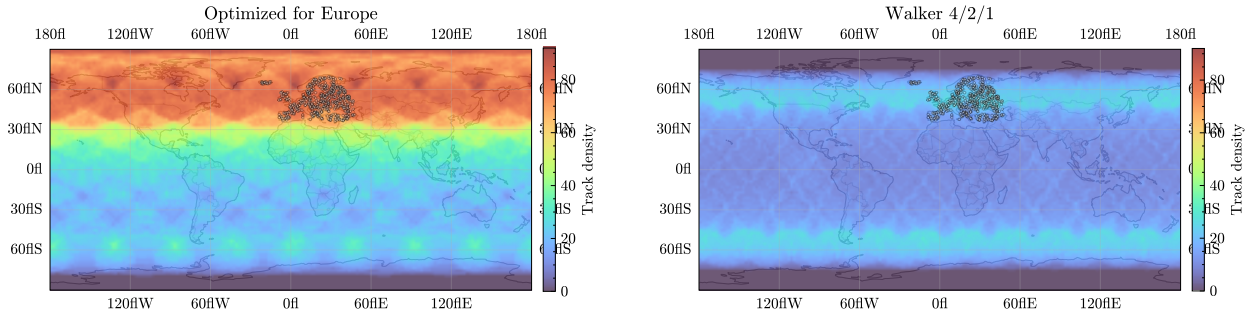
[4] Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

[5] Nathaniel Choo, Darryl Ahner, and Bryan Little. A survey of orbit design and selection methodologies. *The Journal of the Astronautical Sciences*, 71(4), 2024.

[6] Filipe de Avila Belbute-Peres, Kevin Smith, Kelsey Allen, Josh Tenenbaum, and J. Zico Kolter. A differentiable physics engine for deep learning in robotics. *Frontiers in Neurorobotics*, 2019.

[7] Juan A. Fraire, Santiago Henn, Fabio Dovis, Roberto Garello, and Giorgio Taricco. Sparse satellite constellation design for LoRa-based direct-to-satellite Internet of Things. In *IEEE Global Communications Conference (GLOBECOM)*. IEEE, 2020.

[8] Wulfram Gerstner and Werner M. Kistler. *Spiking Neuron Models: Single Neurons, Populations, Plasticity*. Cambridge University Press, 2002.



(a) Optimized for Europe

(b) Walker 4/2/1 baseline

Figure 8: Visibility density over a $24\,\mathrm{h}$ window: number of timesteps each ground bin is within 10° elevation of any satellite. **(a)** The optimized constellation concentrates coverage over European latitudes via eccentric orbits with $\omega \approx 270^\circ$ (apogee dwell in the north). Longitudinal density variations reflect the near-13:1 resonant ground track. **(b)** The Walker baseline distributes coverage symmetrically north/south.

[9] Yuanming Hu, Luke Anderson, Tzu-Mao Li, Qi Sun, Nathan Carr, Jonathan Ragan-Kelley, and Frédo Durand. DiffTaichi: Differentiable programming for physical simulations. In *International Conference on Learning Representations*, 2020.

[10] John T. Hwang, Dae Young Lee, James W. Cutler, and Joaquim R. R. A. Martins. Large-scale multidisciplinary optimization of a small satellite's design and operation. *Journal of Spacecraft and Rockets*, 51(5):1648–1663, 2014.

[11] Eric Jang, Shixiang Gu, and Ben Poole. Categorical reparameterization with Gumbel-softmax. *International Conference on Learning Representations*, 2017.

[12] Nitin Kamra and Yan Liu. Gradient-based optimization for multi-resource spatial coverage problems. In *Proceedings of the 37th Conference on Uncertainty in Artificial Intelligence*, volume 161 of *PMLR*, 2021.

[13] Hao Li, Zheng Xu, Gavin Taylor, Christoph Studer, and Tom Goldstein. Visualizing the loss landscape of neural nets. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[14] Daniele Mortari, Matthew P. Wilkins, and Christian Bruccoleri. The flower constellations. *Journal of the Astronautical Sciences*, 52(1–2):107–127, 2004.

[15] Jack Naylor, Raghav Mishra, Nicholas H. Barbara, and Donald G. Dansereau. dLITE: Differentiable lighting-informed trajectory evaluation for on-orbit inspection. *arXiv preprint arXiv:2512.16011*, 2025.

[16] Sung Wook Paek, Sangtae Kim, and Olivier de Weck. Optimization of reconfigurable satellite constellations using simulated annealing and genetic algorithm. *Sensors*, 19(4):765, 2019.

[17] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32, 2019.

[18] Judea Pearl. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, 1988.

[19] David A. Vallado, Paul Crawford, Richard Hujsak, and T. S. Kelso. Revisiting spacetrack report #3. *AIAA/AAS Astrodynamics Specialist Conference and Exhibit*, 2006.

[20] John G. Walker. Satellite constellations. *Journal of the British Interplanetary Society*, 37:559–572, 1984.

[21] David O. Williams Rogers, Dongshik Won, Dongwook Koh, Kyungwoo Hong, and Hang Woon Lee. Optimal satellite constellation configuration design: A collection of mixed integer linear programs. *Journal of Spacecraft and Rockets*, 2026. doi:10.2514/1.A36518.

A Hyperparameter Tuning

This appendix provides the complete experimental setup and raw results for hyperparameter tuning described in §4.2.

Constellation configuration. The tuning uses a 24-satellite constellation (6×4) at $550\,\mathrm{km}$ altitude with eccentricity ≈ 0, inclination fixed at $60°$, and RAAN and mean anomaly as free parameters. RAAN is shared within each plane, giving 30 effective degrees of freedom (6 unique RAANs + 24 mean anomalies).

Optimization settings. Each of the four initial configurations is optimized for 800 iterations with AdamW ($\eta = 10^{-2}$), propagation over $24\,\mathrm{h}$ discretized into $K = 240$ time steps, a ground grid of $36 \times 72 = 2592$ points spanning $\pm 70°$ latitude with $\cos(\mathrm{lat})$ area weights, minimum elevation $\alpha_{\min} = 10°$, sigmoid temperature $\tau = 2°$, LogSumExp temperature $\beta = 10\,\mathrm{min}$, revisit weight $\lambda = 0.1$, mean revisit reduction, and GMST randomization enabled. The optimization uses the *default* relaxation parameters, not the calibrated ones—the goal is to obtain four diverse final solutions whose relative quality is known from the hard metrics.

Initial RAAN configurations. The four configurations are: near-uniform $[0, 60, 120, 180, 240, 300]°$, moderate $[0, 30, 120, 200, 210, 300]°$, clustered $[0, 10, 20, 30, 40, 50]°$, and two-cluster $[0, 5, 180, 185, 270, 275]°$. All use the same random mean anomaly initialization (NumPy `RandomState(42)`, uniform on $[0, 360)°$).

Grid search parameters. The grid search evaluates the relaxed loss of all four final solutions under each combination of: coverage softness $\tau_{\text{cov}} \in \{1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0\}°$, revisit softness $\tau_{\text{rev}} \in \{0.1, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0\}°$, LogSumExp temperature $\beta \in \{3.0, 5.0, 7.5, 10.0, 15.0, 20.0\}\mathrm{min}$, and revisit weight $\lambda \in \{0.05, 0.1, 0.2, 0.5, 1.0, 2.0\}$. This gives $7 \times 7 \times 6 \times 6 = 1764$ combinations, each requiring 4 gradient-free forward passes (one per solution), for 7056 evaluations total.

Validity criterion. A combination is *valid* if the relaxed loss correctly ranks the good solutions below the bad:

$$\max\big(\mathcal{L}_{\text{near-uniform}},\, \mathcal{L}_{\text{moderate}}\big) < \min\big(\mathcal{L}_{\text{clustered}},\, \mathcal{L}_{\text{two-cluster}}\big). \tag{16}$$

The *margin* is the difference between the right- and left-hand sides. Among valid combinations, we recommend the one that maximizes τ_{cov} (smoothest coverage gradients), then τ_{rev}, with margin as tiebreaker.

Reproducibility. The experiment is implemented in `experiments/exp0_hyperparam_tuning.py`. The full list of valid combinations is written to `experiments/exp0_valid_combos.txt`, sorted by descending coverage softness.